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Recall:

Prop 2.1. $\mathbb{Z}$ is a PID: Every ideal I of $\mathbb{Z}$ is principal (of the form $I = \langle m \rangle$).

Bézout's lemma. Let $a, b \in \mathbb{Z}$, not both 0. Write

$d = (a, b)$ and consider $I_{a,b} = \{ax + by : x, y \in \mathbb{Z}\}$. Then

$I_{a,b}$ is an ideal and $I_{a,b} = \langle d \rangle$. Consequently, there

are $x, y \in \mathbb{Z}$ s.t. $ax + by = d$.

Euclid's lemma. Let $a, b \in \mathbb{Z}$ and let p be prime. Then

$$p \mid ab \Rightarrow p \mid a \text{ or } p \mid b.$$

Rmk 3.1. Applying Euclid's lemma repeatedly shows

$$p \mid a_1 a_2 \cdots a_k \Rightarrow p \mid a_i \text{ for some } 1 \leq i \leq k.$$

Moreover, if $a_1, \dots, a_k$ are primes, then

$$p \mid a_1 a_2 \cdots a_k \Rightarrow p = a_i \text{ for some } 1 \leq i \leq k.$$

Pf of Uniqueness Part of FTA: Let $n > 1$.

① n is prime: $n = n \cdot 1 = 1 \cdot n \quad \checkmark$

② n is composite: Suppose n is minimal s.t.

$$n = p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_l \quad (k \geq l)$$

are distinct factorizations. Since $p_1 \mid q_1 \cdots q_l$, Euclid's lemma $\Rightarrow$

p_1 divides some q_j . But q_j is prime. So $p_1 = q_j$.

Rearrange $g_1, \dots, g_t$ s.t. $g_j = g_1$. So $p_1 = g_1$.

$$n = p_1 p_2 \dots p_k = p_1 g_2 \dots g_t$$

$$\Rightarrow \frac{n}{p_1} = p_2 \dots p_k = g_2 \dots g_t.$$

Remember n is minimal and $1 < \frac{n}{p_1} < n$. A contradiction. $\square$

FTA can be restated as:

Thm (UF in $\mathbb{Z}$) Every nonzero, nonunit $n \in \mathbb{Z}$ can be written as

a product of primes and units, uniquely up to the order

of factors and multiplication by units.

Note: A domain with such a property is called a unique factorization

domain (UFD).

Observation: The same proof shows that a domain equipped with a DA (called Euclidean domain) is both a PID and a UFD.

Thm. $\text{PID} \Rightarrow \text{UFD}$.

How to find $\gcd(a, b)$?

Method 1. List all common divisors and take the largest. 😞

Method 2. Factorize a, b and combine. 😞 Factorizing is hard!

$$\begin{aligned} \text{E.g. } a &= 63 = 3 \cdot 3 \cdot 7 = 3^2 \cdot 7 \\ b &= 42 = 2 \cdot 3 \cdot 7 \end{aligned} \Rightarrow (a, b) = 3 \cdot 7 = 21.$$

Method 3. Euclidean algorithm. 😊

$$\begin{aligned} I_{a,b} &= \{ax + by : x, y \in \mathbb{Z}\} = I_{a, a-b} \\ &= I_{a, a-2b} \\ &= \dots \end{aligned}$$

By DA, $a = q \cdot b + r$, $0 \leq r < |b|$. Then

$$I_{a,b} = I_{a, a - q \cdot b} = I_{a,r} \quad (q \neq 0).$$

keep going for a, r until the remainder becomes 0.

E.g. Find $(1533, 1225)$.

$$1533 = 1225 \cdot 1 + 308$$

$$1225 = \underbrace{308 \cdot 3}_{924} + 301$$

$$308 = 301 \cdot 1 + 7$$

$$301 = 7 \cdot 43 + 0$$

$\uparrow$
gcd

Congruences.

Defn (Congruence) Let $a, b, m \in \mathbb{Z}$ with $m \neq 0$. We say

a is congruent to b modulo m if $m \mid (a-b)$.

Notation. $a \equiv b \pmod{m}$

" $\equiv$ " is an equivalence relation:

(1) Reflexivity: $a \equiv a \pmod{m}$.

(2) Symmetry: $a \equiv b \pmod{m} \Leftrightarrow b \equiv a \pmod{m}$

(3) Transitivity: $a \equiv b \pmod{m}, b \equiv c \pmod{m}$

$$\Rightarrow a \equiv c \pmod{m}.$$

Properties :

$$1. \quad a_1 \equiv b_1 \pmod{m} \quad , \quad a_2 \equiv b_2 \pmod{m}$$

$$\Rightarrow \begin{cases} a_1 \pm a_2 \equiv b_1 \pm b_2 \pmod{m} \\ a_1 a_2 \equiv b_1 b_2 \pmod{m} \end{cases}$$

Special cases : ① $a \equiv b \pmod{m} \Leftrightarrow a \pm c \equiv b \pm c \pmod{m}$.

$$\textcircled{2} \quad a \equiv b \pmod{m} \Rightarrow ka \equiv kb \pmod{m}$$

$$\textcircled{3} \quad a \equiv b \pmod{m} \Rightarrow a^k \equiv b^k \pmod{m}, \quad k \geq 0.$$

$$2. \quad a \equiv b \pmod{mn} \Rightarrow a \equiv b \pmod{m}, \quad a \equiv b \pmod{n}.$$

$$3. \quad ka \equiv kb \pmod{m} \Leftrightarrow \frac{m}{d} \cdot a \equiv \frac{m}{d} \cdot b \pmod{\frac{m}{d}} \quad (d = (k, m))$$

($k \neq 0$)

$$\Leftrightarrow a \equiv b \pmod{\frac{m}{d}}$$

Warning : $2 \cdot 2 \equiv 2 \pmod{2}$, but $2 \not\equiv 1 \pmod{2}$.